

1. Theme gallery

Every panel pairs executable settings with the rendering produced by this checkout. Swatches show the unlit OBJ material colors. Compilation also checks theme metadata, expected colors, and the no-op controls.

The samples use published structural data: RCSB PDB 1HHO and 1CRN, AlphaFold human insulin precursor AF-P01308-F1 (model v6), the official SWISS-MODEL QMEAN example model_001, and Mol*'s charge-annotated 7QPD. The prediction scores and computed charges are genuine published annotations, not experimental measurements or manually assigned test values. See `data/README.md` for source URLs, licenses, and the reproducible PDB + QMEAN JSON to CIF conversion. Source coordinates and scores are preserved.

The QMEAN data and its rendered panel are adapted from SWISS-MODEL, Computational Structural Biology Group, SIB / University of Basel, under CC BY-SA 4.0. QMEAN: Benkert et al. (2011), doi:10.1093/bioinformatics/btq662. AlphaFold: Jumper et al. (2021), doi:10.1038/s41586-021-03819-2, data under CC BY 4.0.

Use the following setup once, then append any panel's render call. For checkout-local debugging, replace the package import with `#import "../lib.typ" as molfig`.

```
#import "@preview/molfig:0.1.4"

// Source files and attribution: examples/data/README.md.
#let view = (
  mesh-format: "obj",
  quality: "medium",
  center: true,
  output-format: "png",
  config: (
    azimuth: 35,
    elevation: 24,
    projection: "orthographic",
    width: 768,
    height: 648,
    background: "",
  ),
  width: 64mm,
  height: 54mm,
)
```

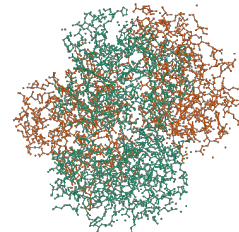
2. All seven global themes

An explicit `globalName` replaces every component theme. In particular, `globalName: "chain-id"` forces chain coloring; the default `color-theme: "chain-id"` instead preserves the representation preset. Structural themes use ball-and-stick; per-residue annotation themes use cartoon. Both are selected explicitly to avoid `ViewerAuto` annotation selection.

Chain ID

```
#molfig.render(  
  read("data/1HH0.cif", encoding: none),  
  ..view,  
  format: "cif",  
  representation: "ball-and-stick",  
  assembly: "1",  
  theme: (  
    globalName: "chain-id",  
  ),  
)
```

1HHO oxyhemoglobin: alpha and beta chains have distinct colors, repeated in the second assembly copy.

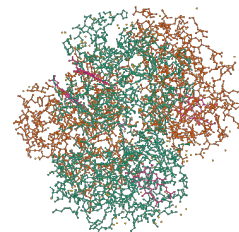


2 material colors

Entity ID

```
#molfig.render(  
  read("data/1HH0.cif", encoding: none),  
  ..view,  
  format: "cif",  
  representation: "ball-and-stick",  
  assembly: "1",  
  theme: (  
    globalName: "entity-id",  
  ),  
)
```

1HHO: alpha/beta proteins, heme, oxygen, phosphate, and water are separate entities. Copies retain entity colors.

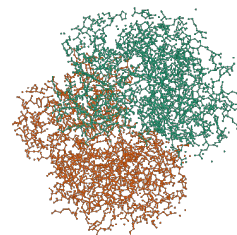


6 material colors

Operator name

```
#molfig.render(  
  read("data/1HH0.cif", encoding: none),  
  ..view,  
  format: "cif",  
  representation: "ball-and-stick",  
  assembly: "1",  
  theme: (  
    globalName: "operator-name",  
  ),  
)
```

1HHO assembly 1: the identity and twofold-related copies have different colors, regardless of chain or entity.

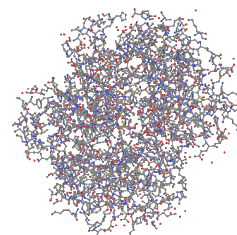


2 material colors

Element symbol

```
#molfig.render(  
  read("data/1HH0.cif", encoding: none),  
  ..view,  
  format: "cif",  
  representation: "ball-and-stick",  
  assembly: "1",  
  theme: (  
    globalName: "element-symbol",  
    carbonColor: "element-symbol",  
  ),  
)
```

1HHO: carbon is grey, nitrogen blue, and oxygen red. Heme iron, sulfur, and phosphorus keep their element colors.



6 material colors

pLDDT confidence

```
#molfig.render(  
  read("data/AF-P01308-F1-model_v6.cif", encoding: none),  
  ..view,  
  format: "cif",  
  representation: "cartoon",  
  assembly: "asymmetric-unit",  
  theme: (  
    globalName: "plddt-confidence",  
  ),  
)
```

AlphaFold human insulin precursor, model v6: genuine predicted pLDDT spans three confidence bands; no residue exceeds 90.

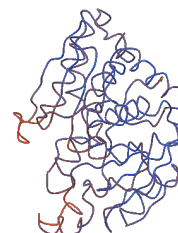


3 material colors

QMEAN score

```
#molfig.render(  
  read("data/qmean-model_001.cif", encoding: none),  
  ..view,  
  format: "cif",  
  representation: "cartoon",  
  assembly: "asymmetric-unit",  
  theme: (  
    globalName: "qmean-score",  
  ),  
)
```

Official SWISS-MODEL QMEAN example model_001. Published local scores are mapped to their original residues, without rounding or rescaling.

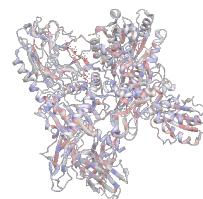


142 material colors (12 shown)

SB-NCBR partial charges

```
#molfig.render(  
  read("data/7qpd.fw2.cif", encoding: none),  
  ..view,  
  format: "cif",  
  representation: "cartoon",  
  assembly: "asymmetric-unit",  
  theme: (  
    globalName: "sb-ncbr-partial-charges",  
  ),  
)
```

Mol*'s 7QPD example: computed SQE+qp/Schindler 2021 atomic charges, summed per residue. Negative is red; positive is blue.



92 material colors (12 shown)

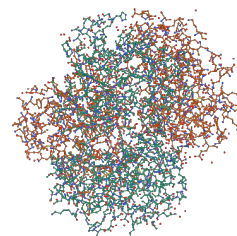
3. All three carbon subthemes

These calls set carbonColor alone. Carbon changes; nitrogen and oxygen must not. No globalName or color-theme is supplied.

Carbon: chain ID

```
#molfig.render(  
  read("data/1HH0.cif", encoding: none),  
  ..view,  
  format: "cif",  
  representation: "ball-and-stick",  
  assembly: "1",  
  theme: (  
    carbonColor: "chain-id",  
  ),  
)
```

Only carbon follows its chain. Nitrogen and oxygen keep their element colors. No global theme is set.

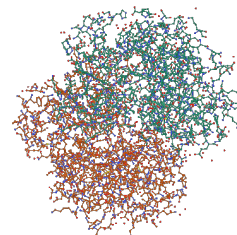


7 material colors

Carbon: operator name

```
#molfig.render(  
  read("data/1HH0.cif", encoding: none),  
  ..view,  
  format: "cif",  
  representation: "ball-and-stick",  
  assembly: "1",  
  theme: (  
    carbonColor: "operator-name",  
  ),  
)
```

Only carbon distinguishes the two assembly copies. This exercises carbonColor without globalName.

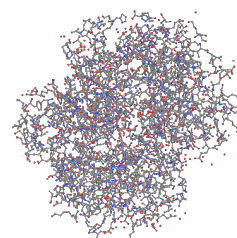


7 material colors

Carbon: element symbol

```
#molfig.render(  
  read("data/1HH0.cif", encoding: none),  
  ..view,  
  format: "cif",  
  representation: "ball-and-stick",  
  assembly: "1",  
  theme: (  
    carbonColor: "element-symbol",  
  ),  
)
```

Carbon becomes grey using only carbonColor; nitrogen and oxygen are unchanged.



6 material colors

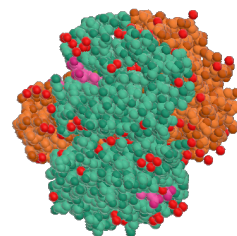
4. Spacefill preset control

Carbon subthemes apply only within element-symbol coloring. The first panel must match spacefill with an empty theme dictionary; the second explicitly replaces its illustrative preset.

Spacefill: preset retained

```
#molfig.render(  
  read("data/1HH0.cif", encoding: none),  
  ..view,  
  format: "cif",  
  representation: "spacefill",  
  assembly: "1",  
  theme: (  
    carbonColor: "element-symbol",  
  ),  
)
```

A carbon subtheme alone does not replace the illustrative spacefill preset. Carbon stays lightened by entity.

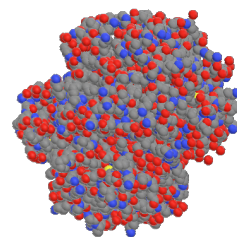


9 material colors

Spacefill: explicit element theme

```
#molfig.render(  
  read("data/1HH0.cif", encoding: none),  
  ..view,  
  format: "cif",  
  representation: "spacefill",  
  assembly: "1",  
  theme: (  
    globalName: "element-symbol",  
    carbonColor: "element-symbol",  
  ),  
)
```

Select element-symbol explicitly to obtain grey carbon in spacefill. Geometry and camera are unchanged.



6 material colors

5. Assembly versus crystal symmetry

All three panels must match. The public API currently cannot generate crystal contacts or crystallographic symmetry expansions; these are negative controls for biological assemblies, not demonstrations of a crystal-symmetry scene.

These panels reuse the bundled RCSB PDB 1CRN biological assembly 1. The build checks identical geometry and material assignments, not only similar images. No crystal-symmetry scene is implied by this control.

Assembly: baseline

```
#molfig.render(  
  read("data/1crn.bcif", encoding: none),  
  ..view,  
  format: "bcif",  
  representation: "cartoon",  
  assembly: "1",  
  theme: (:),  
)
```

RCSB PDB 1CRN, assembly 1: cartoon geometry with its preset chain color.



1 material color

Assembly: symmetryColor only

```
#molfig.render(  
  read("data/1crn.bcif", encoding: none),  
  ..view,  
  format: "bcif",  
  representation: "cartoon",  
  assembly: "1",  
  theme: (  
    symmetryColor: "operator-name",  
  ),  
)
```

The rendering must equal the baseline. Biological assembly operators are not crystal-symmetry operators.



1 material color

Assembly: global + symmetry

```
#molfig.render(  
  read("data/lcrn.bcif", encoding: none),  
  ..view,  
  format: "bcif",  
  representation: "cartoon",  
  assembly: "1",  
  theme: (  
    globalName: "chain-id",  
    symmetryColor: "qmean-score",  
  ),  
)
```

The rendering must still equal the baseline. An inapplicable symmetry override must not replace the chain theme with missing-score grey.



1 material color